Nothing Beats Zero: Detectability Bounds for a Leakage-Controlled, Search-Corrected Borsa Istanbul Forecasting Benchmark

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Abstract

Short-horizon equity forecasting studies usually report a model that beats a benchmark. This one reports the apparatus that would have detected such a model, the finding that none exists in the data examined, and—the part that is usually missing—a quantitative account of what that finding is worth. On a fixed four-stock Borsa Istanbul prototype universe over 251 sessions, seven forecasters are fitted under a date-grouped purged walk-forward protocol with an executable next-open-to-close target, and every out-of-sample prediction is persisted. The 480 out-of-sample rows are shown to carry roughly 177 independent observations because same-session returns correlate at 0.570; loss differentials are therefore aggregated to one value per session, which multiplies the resulting p-values by a median factor of 52. Diebold–Mariano tests with the Harvey–Leybourne–Newbold correction under Holm family-wise control find four of six fitted models significantly *worse* than a zero-return null and none better, Hansen's test of superior predictive ability does not reject, and across a 72-configuration grid of the fold geometry and portfolio breadth the null attains the best out-of-sample R^2 in all 72. We then invert the same apparatus. Three results follow, each computed from the run's own artifacts. First, the design could only have separated an out-of-sample R^2 above 0.113 from zero at conventional size and power, roughly eleven times the largest effect this literature credibly reports; reaching 0.01 would take $15{,}126$ sessions. Second, widening the cross-section cannot fix this: at a mean pairwise correlation of $\bar{\rho}$ a session carries at most $1/\bar{\rho}$ independent observations however many names are added, worth a 9% standard-error gain here for an unlimited universe. Third, a closed-form feasibility bound shows the cost schedule demands a cross-sectional information coefficient of 0.310 from any forecaster in this design, against the 0.039 achieved—and no universe width restores feasibility at that level of skill. The negative result is therefore a statement about the experiment, not about the market, and we give the sample sizes, breadths and cost levels at which the question becomes answerable. The committed run replays byte-for-byte from bundled inputs, and sixteen seeded defects in the inference code are each required to break a guarding test.

1 Introduction

A short-horizon equity forecasting result is easy to manufacture and hard to verify. The standard failure modes are well catalogued: a target the backtest could not have traded, a validation split that lets a model see its own future, a metric whose null is never stated, a p-value computed on rows that are not independent, and a headline drawn from whichever configuration happened to look best. Each inflates apparent performance without leaving an obvious trace in the code.

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There is a second failure mode, less discussed, that applies with equal force to the studies that report nothing. A negative result carries information only in proportion to the power of the design that produced it. "We tested and found no effect" and "we ran a test that could not have found the effect even had it been there" are different claims, and in this literature they are almost never distinguished, because the second requires computing something the first does not. Gelman and Carlin [1] make the general argument: a design should be characterised by what it can resolve before its output is interpreted. Button et al. [2] show what happens to a field that does not do this.

This work does both halves. The first half is an evaluation apparatus strict enough that a real effect would survive it, applied to a Borsa Istanbul panel; its verdict is negative at every level tested. The second half turns that same apparatus around and asks what it was capable of detecting. The answer reframes the result entirely: the design is not evidence that short-horizon prediction on this market is impossible, it is evidence that the question cannot be settled at this sample size, this cross-sectional breadth, or this cost level—and we compute how far each would have to move.

Contributions

C1. A dependence-aware evaluation protocol for panel forecast comparison. Four tickers observed on one session are not four independent observations. Section 4.4 measures the inflation and changes the unit of inference to the session rather than correcting after the fact; the row-level statistics are retained and reported alongside, so the size of the error the usual practice makes is visible rather than argued about. Here it is a median factor of 52 in the p-values.

C2. Detectability bounds for the design. Inverting the accepted tests at conventional size and power yields the smallest out-of-sample R^2 and the smallest Sharpe ratio the experiment could have separated from its null (Section 5). Both exceed anything the literature credibly reports, by an order of magnitude.

C3. An information ceiling for correlated panels (Proposition 1). Under equicorrelation, one session of a k-name cross-section carries $k/(1 + (k - 1)\bar{\rho})$ independent observations, which is bounded above by $1/\bar{\rho}$. Breadth therefore cannot substitute for length: past a few dozen names, adding more buys essentially no statistical precision. On this panel an unlimited universe would improve standard errors by 9%.

C4. A breadth–cost feasibility bound (Proposition 2). For a long-only rule holding the top k of N ranked names against a round-trip cost c and a target of volatility σ, positive expected net return requires a cross-sectional information coefficient above $c/(\sigma\,\lambda(N, k))$, where λ is the mean of the top-k standard normal order statistics. The bound is a property of the experimental design, computable before any model is fitted, and it rules this design out by a factor of eight.

C5. An effective trial count for correlated searches (Proposition 3). The False Strategy Theorem assumes independent trials; a configuration grid reuses most of the same sessions and disperses less. Equating the two thresholds recovers the number of independent searches the grid behaves like—here 7.01 of 72—and brackets the search correction between its two defensible readings. The grid maximum falls below both.

C6. A verified evaluator. The inference code is itself subject to mutation testing: sixteen real defects are reintroduced one at a time and each is required to break a specific test (Appendix F). Three tests were found to be decorative this way and repaired. The committed run replays byte-for-byte from bundled inputs, and every table and figure in this manuscript is generated from that bundle by a tested module.

The outcome is negative and is retained because methodological validity, not model novelty, is the acceptance criterion.

2 Related work

Forecast comparison. Diebold and Mariano [3] give the standard test of equal predictive accuracy; Harvey et al. [4] show the statistic is oversized in small samples and supply the correction used here. Diebold [5] later emphasised that the test compares forecasts rather than the models producing them, which is the interpretation adopted below.

Data snooping. White [6] shows that selecting the best of m models and testing that model alone is invalid, and gives a bootstrap Reality Check for the joint null. Hansen [7] studentizes the statistic and recentres the bootstrap. Sullivan et al. [8] apply the Reality Check to a century of technical trading rules and find the apparent performance of the best rule largely disappears once the size of the search is accounted for; that result is the direct template for Section 6.2.

Dependent-data resampling. Politis and Romano [9] introduce the stationary bootstrap; Politis and White [10], corrected by Patton et al. [11], give the automatic block-length rule used here.

Backtest overfitting. Lo [12] derives the sampling distribution of the Sharpe ratio and shows the square-root annualisation rule is wrong under autocorrelation. Bailey and López de Prado [13] convert the Sharpe ratio into a probability accounting for skewness and kurtosis, and Bailey and López de Prado [14] deflate it by the number of configurations tried. Bailey et al. [15] show how quickly an unrecorded search produces a spurious backtest, and Harvey et al. [16] make the same argument for the cross-section of expected returns. What none of this literature supplies, and what Section 5.4 adds, is a way to say how many *independent* searches a correlated grid amounts to.

Effect sizes worth detecting. Welch and Goyal [17] document that most equity-premium predictors fail out of sample; Campbell and Thompson [18] argue that an out-of-sample R^2 of well under one percent is already economically meaningful, which is the scale against which Section 5 measures this design. Gu et al. [19] report out-of-sample R^2 in the same range for machine-learning models on a monthly US cross-section far larger than this one. Any short-horizon study claiming an R^2 an order of magnitude above that range is more likely reporting a leak than a discovery.

Design analysis. Cohen [20] is the standard reference for inverting a test at a stated power. Gelman and Carlin [1] argue that this should be done with a plausible effect size supplied externally rather than with the observed estimate, which is the procedure of Section 5; Button et al. [2] and Ioannidis [21] document the consequences of skipping it. To our knowledge this is not standard practice in the empirical finance forecasting literature, where negative results are reported without any accompanying statement of what the design could have resolved.

Breadth and costs. Grinold [22] relates the information ratio of an active strategy to the product of skill and the square root of breadth. Proposition 2 is a cost-aware statement in the same spirit but with the opposite use: rather than predicting attainable performance from assumed skill, it converts an observed cost schedule into the skill a design requires, using the order-statistic machinery of David and Nagaraja [23]. Novy-Marx and Velikov [24] document that many published anomalies do not survive realistic trading costs; Section 6.3 reproduces that pattern in miniature, and Section 5.3 explains why it was inevitable here.

Borsa Istanbul. Kara et al. [25] report high directional accuracy for neural networks and support vector machines on the Istanbul Stock Exchange index. That study, like most in the genre, reports directional accuracy at index level without transaction costs, without a stated null, and without a correction for the number of specifications examined. The present work does not contradict it; it measures a harder quantity on a different universe and reports what survives when those three things are supplied.

3 Data

The committed run contains 1,004 provider rows for GARAN, ISCTR, KCHOL and THYAO from 2025-04-07 to 2026-04-03. Every execution open and every volume observation is flagged `observed`; proxy opens are rejected, because a proxy open cannot be traded at. Each record carries provider name, provider symbol, source record identifier, retrieval timestamp, and separate open- and volume-quality flags, and the canonical record keeps raw tradable, split-adjusted and total-return price representations separately.

The runner validates expected sessions, duplicates, weekend rows, holidays, timezone, and full-day and half-day session bounds against the committed Borsa Istanbul schedule. The input bundle carries five sourced cash-dividend events and a per-ticker corporate-action query-coverage artifact, so "no action" is an assertion with a source rather than an absence of data. Because the accepted strategy holds for one session and carries no overnight entitlement, those five events are persisted as `no_entitlement` rather than invented income. Split, bonus-issue, ticker-change, rights-issue and delisting handlers exist and are exercised by synthetic tests; the rights-issue and unsourced-delisting paths fail closed rather than guess a policy.

The universe is fixed by construction and therefore free of survivorship bias within its own definition. It is *not* point-in-time index membership, and no claim about BIST-100 is made anywhere in this document.

4 Method

4.1 Executable target

For ticker i and feature date t, only information available after the official close of session t is used. With O and C the raw tradable open and close,

$$r^{\mathrm{exec}}_{i,t+1} = \frac{C_{i,t+1}}{O_{i,t+1}} - 1, \qquad y_{i,t+1} = \mathbf{1}\left\{r^{\mathrm{exec}}_{i,t+1} > 0\right\}. \tag{1}$$

Every panel row stores the feature, execution and target timestamps, and the chronology $t_{\mathrm{feature}} < t_{\mathrm{exec}} \leq t_{\mathrm{start}} < t_{\mathrm{end}}$ is enforced as an invariant rather than assumed.

4.2 Features

Nineteen features enter the pooled model, listed in full in Appendix B: returns over one, five and twenty sessions, moving-average and VWAP ratios, an ATR-to-close ratio, realised volatility, a volume z-score, a drawdown, two cross-sectional quantities, and four calendar encodings. Every one is a ratio, a log ratio, a rank or a bounded periodic function.

Raw close, moving-average levels, absolute ATR, raw VWAP, raw volume and OBV are deliberately excluded. In a pooled model, nominal price scale acts as accidental ticker identification: a model that learns "prices near 300" has learned an identity, not a signal, and the leak is invisible in any per-ticker diagnostic. Missing values are preserved with a reason code and imputed by training-fold median with an accompanying indicator column, so absence is a feature rather than a silently filled zero.

4.3 Validation protocol

Folds are expanding windows over unique trading dates. A training date group is purged when any of its samples has a feature timestamp or target end reaching the first validation feature timestamp, so for every fold k,

$$\max_{j \in \mathrm{train}(k)} t^{\mathrm{end}}_j \; < \; \min_{j \in \mathrm{val}(k)} t^{\mathrm{feature}}_j. \tag{2}$$

Tickers on one date are never split across partitions, and preprocessing is fitted on training rows only. The committed run uses 24 minimum training dates, 10 validation dates, a 10-date step and a one-date embargo, producing 12 folds over 120 disjoint evaluation sessions. Figure 1 draws the partition that was executed, read from the run's own fold artifact; Appendix D tabulates its boundaries.

4.4 Effective sample size and the unit of inference

For k cross-sectional units with average pairwise correlation $\bar{\rho}$, the variance of a cross-sectional mean is inflated by

$$\mathrm{VIF} = 1 + (k-1)\bar{\rho}, \qquad n_{\mathrm{eff}} = \frac{n}{\mathrm{VIF}}. \tag{3}$$

Rather than apply a correction after the fact, the evaluation changes the unit of inference: every loss differential is averaged across the tickers present on a date and all tests run on the resulting one-value-per-session series. Row-level statistics are computed anyway and reported alongside, so the size of the inflation is visible rather than argued about. The estimator averages pairwise Pearson correlations with equal weight over unit pairs, skipping pairs with fewer than three shared sessions rather than treating them as zero, and floors the inflation factor at one: a negative average correlation would imply the pooled sample carries *more* information than independent rows, which is not a claim this evaluation is willing to make. Table 1 and Figure 2 give the realised values.

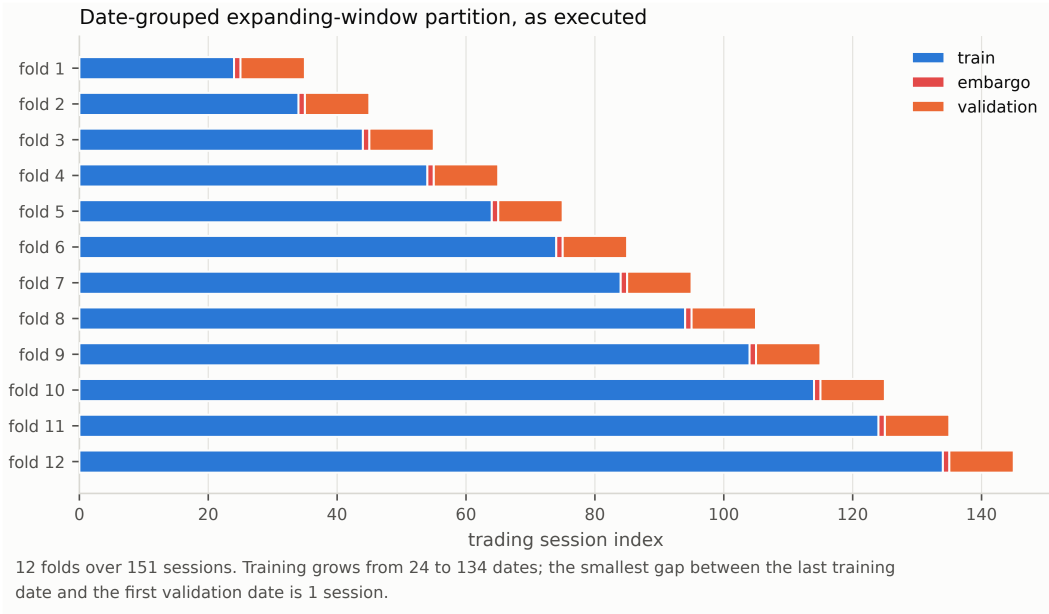

Figure 1: The executed date-grouped expanding-window partition. Nothing is illustrative: each bar spans the dates the fold really used.

Table 1: Cross-sectional dependence in the evaluation panel, and the most an arbitrarily wide cross-section could add at the same correlation.

Quantity	Value
Panel units	4
Evaluated sessions	120
Out-of-sample rows	480
Mean within-session correlation	0.5697
Variance inflation factor	2.7090
Effective independent rows	177.2
Independent rows per session	1.4765
Ceiling on independent rows per session ($1/\bar{\rho}$)	1.7554
Standard-error gain from an unlimited universe	9.0%

4.5 Estimators and metrics

Seven forecasters share the same panel, target, folds, feature manifest, evaluation dates and train-only preprocessing: a zero-return null, a training-fold majority-direction null, a one-session persistence rule, a same-date cross-sectional mean, a rolling training mean, a regularised logistic direction model, and a ridge return model that also ranks the portfolio. The three history heuristics see only targets that had already matured at the prediction timestamp, which is the leakage guard that makes them baselines rather than oracles.

The primary accuracy metric is the zero-mean out-of-sample R^2,

$$R_0^2 = 1 - \frac{\sum_{i,t}(r_{i,t} - \hat{r}_{i,t})^2}{\sum_{i,t} r_{i,t}^2}. \tag{4}$$

The sample-mean version is the wrong choice here: its benchmark uses the mean of the evaluation window, which is not available at prediction time, so it flatters any model that merely learns the window's drift.

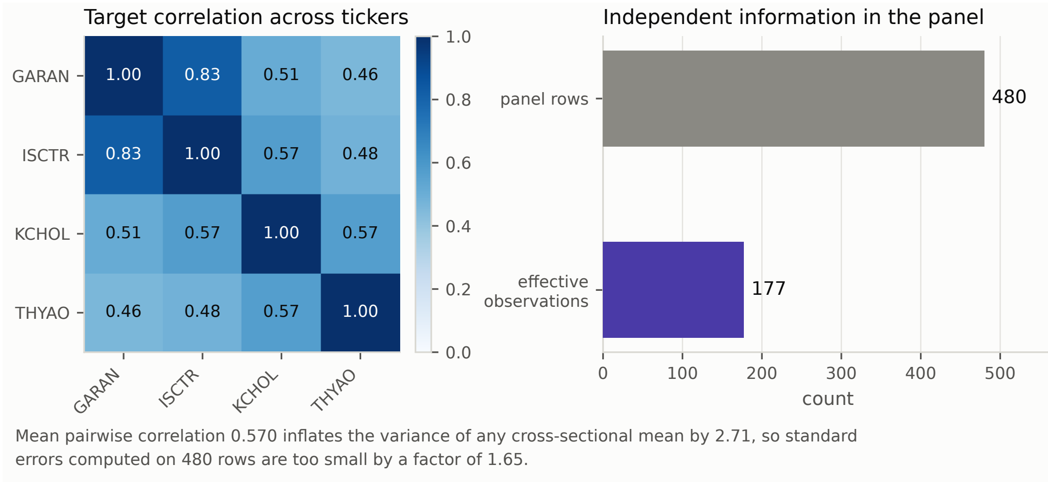

Figure 2: Realised correlation of the executable target across the four tickers, and the sample size it implies.

4.6 Tests

Equal predictive accuracy. With session-aggregated squared-error differentials $d_t = L_t(\text{model}) - L_t(\text{null})$,

$$DM = \frac{\bar{d}}{\sqrt{\hat{\Omega}/n}}, \qquad DM^* = DM\sqrt{\frac{n+1-2h+h(h-1)/n}{n}}, \tag{5}$$

with $\hat{\Omega}$ the Bartlett long-run variance and $h = 1$, referred to a Student t distribution on $n-1$ degrees of freedom [3, 4]. The sign convention is reported with every test: a *positive* statistic means the model is worse than the null. Six models are tested against the same null on the same data, so Holm's step-down procedure [26] controls the family-wise error rate without assuming independence.

Data snooping. With $Z_{k,t} = L_t(\text{null}) - L_t(\text{model } k)$,

$$V_n = \max_k \sqrt{n}\,\bar{Z}_k, \qquad T_n = \max\left[0,\ \max_k \frac{\sqrt{n}\,\bar{Z}_k}{\hat{\omega}_k}\right], \tag{6}$$

whose null distributions come from a stationary bootstrap that resamples every model on the same index draw, preserving cross-model dependence, with the block length chosen automatically [6, 7, 9, 10]. One deviation from Hansen [7] is recorded here because it matters in exactly this regime: the p-values compare untruncated maxima rather than the floored statistic T_n. Applying the floor to both the observed value and every bootstrap draw collapses the comparison onto the atom at zero as soon as no candidate has a positive mean, and returns an arbitrary p-value precisely when the evidence is weakest.

Sharpe ratio under search. For per-period Sharpe $\hat{SR}$ with skewness $\hat{\gamma}_3$ and kurtosis $\hat{\gamma}_4$,

$$\widehat{PSR}(SR^*) = \Phi\left[\frac{(\hat{SR} - SR^*)\sqrt{n-1}}{\sqrt{1 - \hat{\gamma}_3\hat{SR} + \frac{\hat{\gamma}_4 - 1}{4}\hat{SR}^2}}\right], \tag{7}$$

and the deflated Sharpe ratio evaluates it at the threshold the best of N skill-free trials would be expected to reach,

$$SR_0^* = \sqrt{V[\hat{SR}]}\,q(N), \qquad q(N) = (1-\gamma)\,\Phi^{-1}\big(1 - \tfrac{1}{N}\big) + \gamma\,\Phi^{-1}\big(1 - \tfrac{1}{Ne}\big), \tag{8}$$

with γ the Euler–Mascheroni constant [13, 14]. Both quantities are per period; substituting an annualised value inflates the result silently. Lo's autocorrelation-aware annualisation factor [12] is reported alongside the square-root rule, with lags truncated at the Newey–West bandwidth [27, 28] because 120 sessions cannot estimate 251 autocorrelations.

4.7 Portfolio simulation

Long-only, top-k by predicted return, equal weights, next-open execution at observed prices, one-session holding. A name is bought only when its predicted return net of the declared decision-cost assumption is positive. Participation caps and market impact use a 20-session trailing volume reference whose last observation is no later than the signal date. The ledger records signals, orders, fills, positions, cash and costs as separate artifacts and enforces $E_T = E_0 + \mathrm{PnL}_{\mathrm{gross}} + \mathrm{distributions} - \mathrm{costs}$ session by session. Cost sensitivity reuses the same trading decisions across multipliers, so only the bill moves. The complete parameter set is in Appendix C.

5 What the design could have detected

Everything above answers "is there evidence of skill?". This section answers the question those tests cannot: would they have said yes for an effect of the size anyone actually expects? Each quantity below is computed from the run's own artifacts, and each is reported in Table 10, Table 11 or Table 12.

5.1 Minimum detectable accuracy

The statistic in Equation 5 is a mean divided by its standard error. Inverting it at size α and power $1 - \beta$ gives the smallest mean loss differential the design could have separated from zero,

$$\delta_{\min} = \left(t_{1-\alpha/2,\, n-1} + t_{1-\beta,\, n-1}\right) \cdot \mathrm{SE}(\bar{d}), \tag{9}$$

and dividing by the null's mean squared error puts it on the scale of Equation 4. We evaluate this for the candidate with the *smallest* standard error, the one the design had the best chance of separating; quoting any other would flatter the experiment. Following Gelman and Carlin [1] the reference effect is supplied externally rather than taken from the data: we use $R_0^2 = 0.01$, at the upper end of what Campbell and Thompson [18] and Gu et al. [19] treat as economically meaningful out of sample.

Since $\mathrm{SE} \propto n^{-1/2}$, the session count at which a given effect becomes detectable follows by solving Equation 9 for n. The critical values move with n as well, so the solution is found by search rather than by the closed form $n_0(\delta_0/\delta)^2$, which understates the requirement at small n.

The same inversion applies to the portfolio. Setting Equation 7 equal to 0.95 at the search threshold of Equation 8 and solving for $\hat{SR}$ gives the per-session Sharpe ratio that would have counted as evidence of skill after the search actually performed.

Remark 1. The threshold SR_0^* does not shrink with the sample size. Lengthening the record shrinks the sampling noise around $\hat{SR}$ but leaves the bar the search sets exactly where it was, so a strategy whose true Sharpe ratio lies below SR_0^* can never establish skill at any record length. The search correction is a floor on the effect size, not merely a penalty on the p-value.

5.2 The information ceiling of a correlated panel

Widening the cross-section is the usual answer to a short sample. It does not work when the units share a market factor.

Proposition 1 (Panel information ceiling)**.** *Let a session contain k units whose targets are equicorrelated at $\bar{\rho} \in (0, 1)$. Then the session carries*

$$m(k) = \frac{k}{1 + (k-1)\bar{\rho}} \tag{10}$$

independent observations, m is strictly increasing in k, and $m(k) \uparrow 1/\bar{\rho}$ as $k \to \infty$.

The proof is immediate from Equation 3 and is given in Appendix A. The consequence is the part that matters: the gain available from breadth is bounded by $1/(\bar{\rho}\, m(k))$ in variance and by its square root in standard error, so a study whose cross-section is already moderately wide gains almost nothing from widening it further. Only more sessions help. Figure 10 draws the curve and marks where this panel sits on it.

5.3 The cost floor on breadth

Breadth does buy something else: selectivity. Ranking N names and holding the best k concentrates the forecast into its right tail, and the strength of that concentration is the mean of the top-k standard normal order statistics [23],

$$\lambda(N,k) = \frac{1}{k} \sum_{i=N-k+1}^{N} \mathbb{E}\big[Z_{(i:N)}\big] = \frac{N}{k} \int_0^1 \Phi^{-1}(u)\, I_u(N-k,\ k)\,\mathrm{d}u, \tag{11}$$

with I_u the regularised incomplete beta function. The second form follows from exchangeability—a draw is in the top k exactly when at most $k-1$ of the other $N-1$ exceed it—and replaces k quadratures by one.

Proposition 2 (Breadth–cost feasibility)**.** *Let the forecast $\hat{r}$ and the realised return $r \sim \mathcal{N}(0,\sigma^2)$ be jointly normal with correlation ρ. A long-only rule that ranks N names by $\hat{r}$, holds the top k in equal weight for one period, and pays a round-trip cost c on notional, has positive expected net return only if*

$$\rho \ > \ \frac{c}{\sigma\,\lambda(N,k)}. \tag{12}$$

The proof is in Appendix A. Three features of the bound are worth stating plainly. It is *generous*: it ignores estimation error in the ranking, charges the cost only once per round trip, and assumes an unbiased forecast. It is *model-free*: c, σ and $\lambda(N,k)$ are properties of the experimental design, not of the forecaster, so the requirement can be computed before a single model is fitted. And it is *not* a statistical statement: a strategy can fail Equation 12 while producing a forecast that is significantly better than the null, which is exactly what happens when a real but small edge is smaller than the spread.

Equation 12 is a cost-aware relative of the fundamental law of active management [22], used in the opposite direction: rather than predicting attainable performance from assumed skill, it converts an observed cost schedule into the skill a design demands.

5.4 How many searches was the grid?

Equation 8 assumes the N trials are independent draws. A configuration grid is not: neighbouring configurations reuse most of the same sessions, so their Sharpe ratios move together and disperse less than independent trials would. Substituting the *realised* dispersion V_{real} absorbs the dependence but hides its strength.

Proposition 3 (Effective trial count)**.** *Let $V_{\text{ind}} = (1+\hat{SR}^2/2)/n$ be the sampling variance of a single per-period Sharpe ratio [12], and let V_{real} be the realised variance of the N trial Sharpe ratios. Then q in Equation 8 is continuous and strictly increasing on $(1,\infty)$, so*

$$\sqrt{V_{\text{real}}}\; q(N) = \sqrt{V_{\text{ind}}}\; q(N_{\text{eff}}) \tag{13}$$

has a unique solution N_{eff}, the number of independent searches whose expected maximum equals the grid's.

When $V_{\text{real}} < V_{\text{ind}}$ the grid is behaving like fewer independent searches than it contains, and $N_{\text{eff}} < N$. Reporting both thresholds brackets the correction: the independent-trial reading is the conservative one, the realised-dispersion reading is the permissive one, and a grid maximum below both is a conclusion that does not depend on the choice.

6 Results

6.1 No fitted model reaches the null

Every fitted model has negative out-of-sample R^2 (Table 2 and Figure 3). Under Holm correction, four of the six are significantly *worse* than the null and none is better (Table 3); `logistic` and `rolling_mean` are indistinguishable from it. This is a stronger negative statement than "no model beat the benchmark": several models are reliably worse than predicting zero, which is what fitting a linear model to a signal-free panel produces.

Table 2: Out-of-sample accuracy by model, recomputed from the saved prediction rows. The zero-mean R^2 compares against a zero forecast, so the null sits at exactly zero by construction.

Model	n	MAE	RMSE	R^2_0	Spearman IC	Dir. acc.
`zero_return`	480	1.425%	1.898%	0.0000	n/a	53.12%
`rolling_mean`	480	1.461%	1.962%	-0.0680	0.0079	50.42%
`logistic`	480	1.502%	1.971%	-0.0783	-0.0040	48.75%
`ridge`	480	1.651%	2.120%	-0.2470	0.0210	51.04%
`market_direction`	480	1.873%	2.458%	-0.6768	0.0094	52.71%
`majority_direction`	480	1.978%	2.466%	-0.6876	0.1080	53.12%
`previous_return`	480	1.984%	2.634%	-0.9252	0.0324	51.25%

Table 3: Equal predictive accuracy against the zero-return null under squared-error loss. The differential is $\mathrm{loss}(\mathrm{model}) - \mathrm{loss}(\mathrm{null})$, so a positive statistic means the model is worse. The final column repeats each test treating all panel rows as independent draws.

Model	DM^*	p	Holm p	Verdict	Row-level p
`majority_direction`	+5.334	0.0000	0.0000	loses to null	0.0000
`previous_return`	+4.707	0.0000	0.0000	loses to null	0.0000
`market_direction`	+3.824	0.0002	0.0008	loses to null	0.0000
`ridge`	+3.334	0.0011	0.0034	loses to null	0.0000
`logistic`	+1.945	0.0541	0.1081	indistinguishable	0.0015
`rolling_mean`	+1.642	0.1031	0.1081	indistinguishable	0.0103

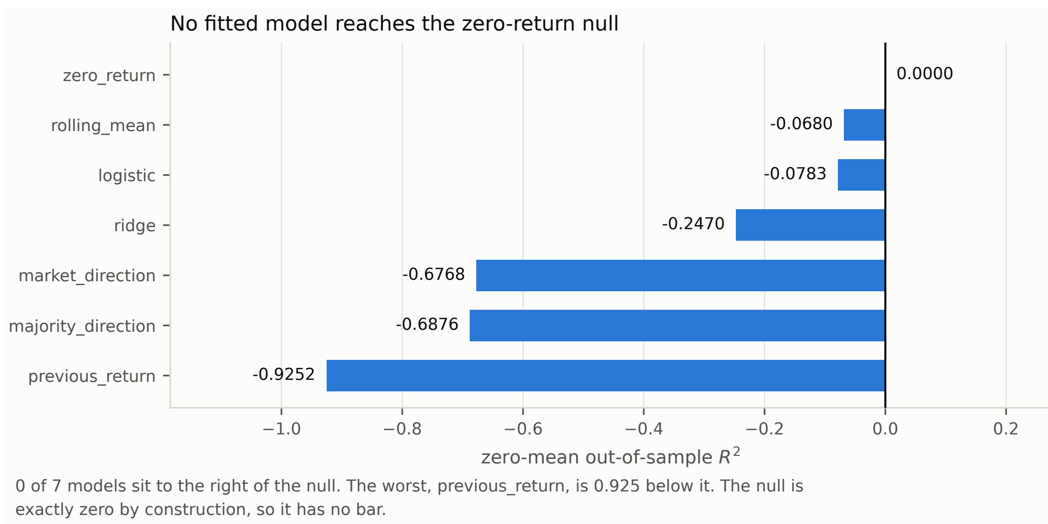

Figure 3: Zero-mean out-of-sample R^2 by model. The null is exactly zero by construction and therefore has no bar.

The right panel of Figure 4 makes the dependence problem concrete. At row level `logistic` reaches $p = 0.0015$; at session level it is 0.0541 and does not survive correction. Nothing about the data changed, only the claim about how much of it is independent. The same reversal applies in the other direction: the four rejections that survive Holm correction are all of the sign that says the model is worse, so the corrected table contains no discovery at all.

Directional accuracy deserves a word, because it is the metric this literature usually leads with. In this window 53.12% of realised targets are non-positive, so a constant "down" predictor attains 53.12% directional accuracy while carrying no information whatever. That is exactly the figure `zero_return` and `majority_direction` reach in Table 2, and it is above what `ridge`, `logistic`

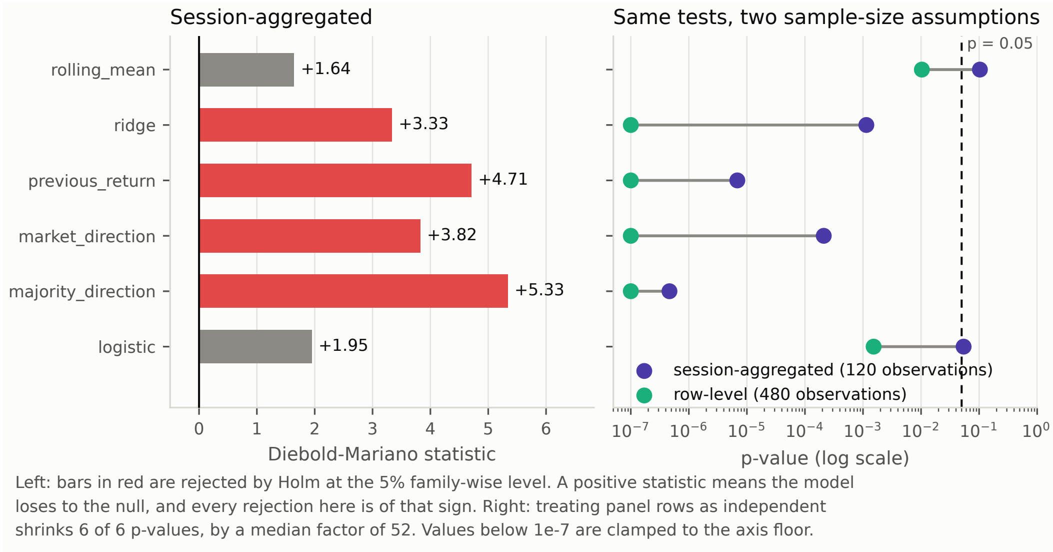

Figure 4: Diebold–Mariano statistics against the zero-return null, and the same six tests under two sample-size assumptions. Treating the 480 panel rows as independent shrinks every p-value, by a median factor of 52.

Table 4: Joint tests that no candidate beats the zero-return benchmark, over 10,000 stationary-bootstrap replications with mean block length 12.26. Hansen's three recentrings bracket the p-value.

Test	Statistic	p
White Reality Check	-0.000268	0.9970
Hansen SPA, lower	0.0000	0.6891
Hansen SPA, consistent	0.0000	0.6891
Hansen SPA, upper	0.0000	1.0000

Table 5: Sharpe-ratio inference. The deflated ratio evaluates the probabilistic Sharpe ratio at the threshold the best of N skill-free configurations would be expected to reach.

Quantity	Value
Evaluated sessions	120
Per-session Sharpe	-0.0284
Annualised, square-root rule	-0.4513
Annualised, Lo autocorrelation-adjusted	-0.4973
Skewness	-0.0722
Kurtosis	7.2290
Probabilistic Sharpe ratio, threshold 0	0.3782
Configurations examined	72
Search threshold	0.1267
Deflated Sharpe ratio	0.0452

and `rolling_mean` manage. Directional accuracy without its base rate is not interpretable, which is why balanced accuracy is reported beside it and neither is used as a headline.

6.2 Nothing survives the search correction

Figure 5 is the data-snooping argument in one picture. The bootstrap null distribution is centred *above* zero because it is the distribution of a maximum over six candidates; that offset is precisely the correction, made visible. The observed maximum lies to the left of it and is itself negative, so the best candidate loses to the null before any correction is applied. Table 4 gives the joint p-values:

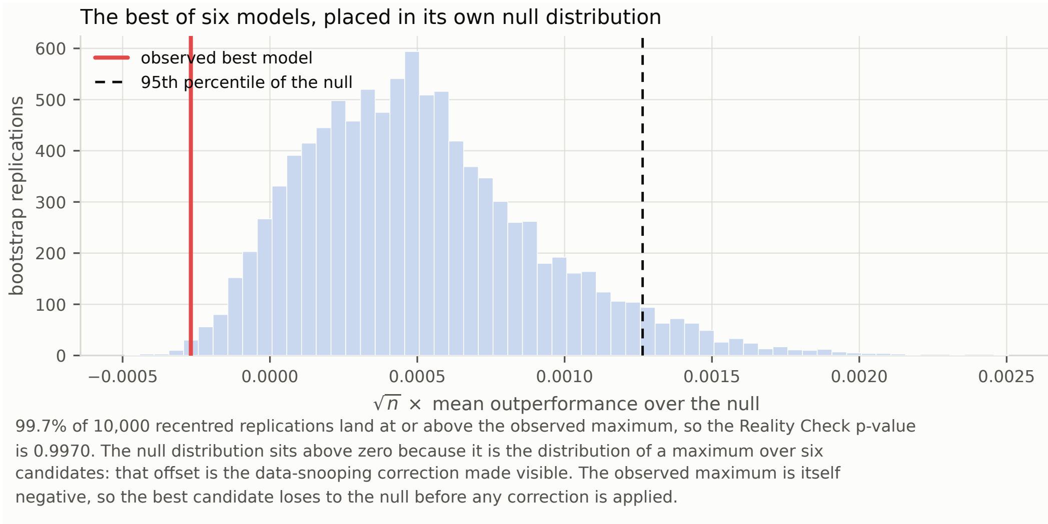

Figure 5: The best of six models placed inside its own bootstrap null distribution. The null is centred above zero because it is the distribution of a maximum over six candidates: that offset is the data-snooping correction made visible. The observed maximum is negative.

Table 6: The configuration grid. Each trial is a complete re-run of the evaluation under one fold geometry and one portfolio breadth.

Quantity	Value
Configurations evaluated	72
Reported configuration rank by Sharpe	16
Median per-session Sharpe	-0.0682
Per-session Sharpe range	-0.1852 to 0.0356
Configurations with positive net return	5.56%
Best configuration net return	4.6259%
Null had the best R_0^2	72 of 72
Expected maximum Sharpe under no skill	0.1267

0.9970 for the Reality Check and 0.6891 for Hansen's consistent recentring, bracketed by 0.6891 and 1.0000. The joint null is nowhere near rejection.

Table 5 carries the same message for the portfolio. Session returns have a kurtosis of 7.2290, so the normal-theory Sharpe interval understates tail risk; accounting for the higher moments, the probability that the true per-session Sharpe exceeds zero is 0.3782, already below even odds before any search correction. Deflating by the 72 configurations examined leaves a deflated Sharpe ratio of 0.0452.

The grid itself varies the minimum training window, the validation width and step, the embargo and the portfolio breadth; each point is a complete re-run of the evaluation, not a re-weighting of a cached result (Table 6, Figure 6, and the full listing in Appendix E). Only 5.56% of those configurations produced a positive net return. The reported configuration ranks 16th of 72 by Sharpe ratio, so it is neither cherry-picked nor unusually unlucky, and the grid maximum of 0.0356 falls short of the 0.1267 that the best of 72 skill-free configurations would be expected to reach. The most compact statement of the result is that across all 72 configurations, the zero-return null attained the best out-of-sample R^2 in 72 of 72.

6.3 Costs, not signal, decide the outcome

The strategy earns a gross return of 7.77% and loses 4.77% of its capital after costs, on $123.7\times$ turnover and TRY $12{,}404$ of modelled cost (Table 7). Because the trading decisions are held fixed

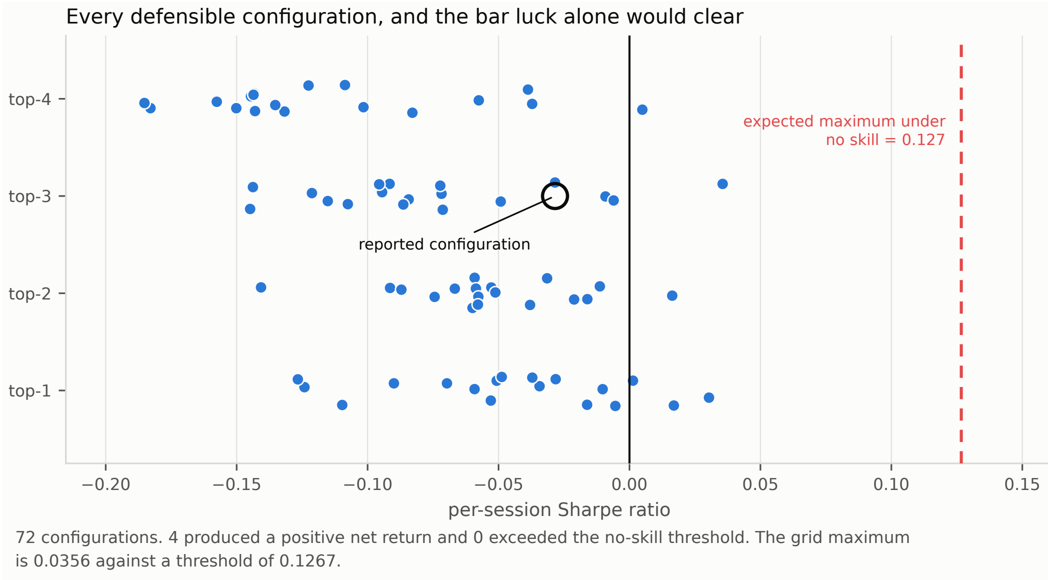

Figure 6: Every configuration in the grid against the Sharpe ratio that skill-free search alone would be expected to produce. The grid maximum does not reach the bar luck sets.

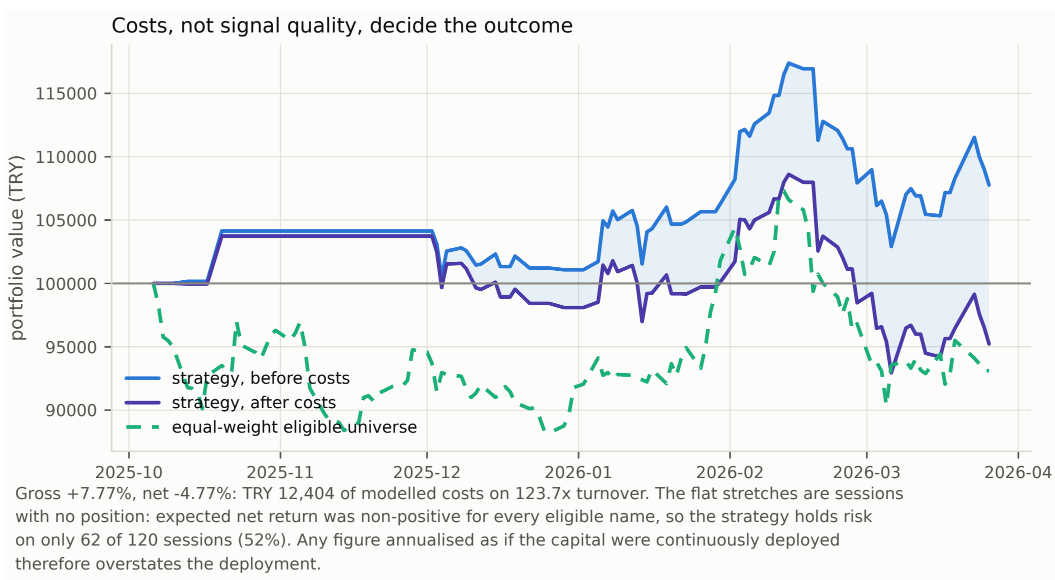

Figure 7: Gross and net equity against the equal-weight eligible universe. The flat stretches are sessions with no position: the strategy holds risk on 62 of 120 sessions, so annualised figures overstate the deployment.

Table 7: The executed long-only top-k portfolio, computed entirely from the persisted event ledgers. Sessions holding risk is reported because annualised figures assume continuous deployment.

Measure	Value
Gross return	7.7694%
Net return	-4.7687%
Annualised return	-9.7521%
Annualised volatility	18.8297%
Sharpe	-0.4513
Maximum drawdown	-14.4181%
Turnover	123.68×
Round trips	143
Sessions holding risk	62 of 120
Equal-weight universe	-5.8309%
Modelled cost	TRY 12,403.77

Table 8: Cost sensitivity. The same trading decisions are reused across the three cases, so only the bill changes; net return cannot improve as costs rise.

Case	Gross	Net	Total cost	Round trips
0.0x	7.7671%	7.7671%	TRY 0.00	143
1.0x	7.7694%	-4.7687%	TRY 12,403.77	143
2.0x	7.7188%	-15.8890%	TRY 23,338.11	143

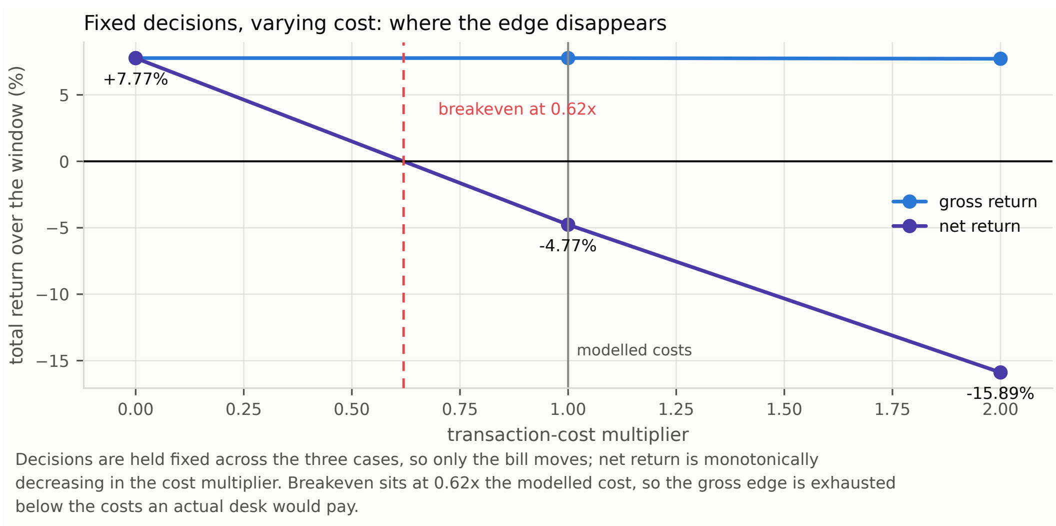

Figure 8: Fixed trading decisions under three cost multipliers. Breakeven sits at $0.62\times$ the modelled schedule.

across the three cost cases, Table 8 isolates the bill: net return falls monotonically, and interpolating between the zero-cost and modelled-cost cases puts breakeven at $0.62\times$ the modelled schedule, comfortably below what an actual desk would pay. This reproduces Novy-Marx and Velikov [24] at small scale: the gross edge is real and the net edge is not.

Figure 7 shows something the return statistics hide. The flat stretches are sessions in which no eligible name had a positive expected net return, so the strategy held nothing; it carries risk on 62 of 120 sessions. Any figure annualised as though the capital were continuously deployed, including the annualised return and Sharpe ratio in Table 7, therefore overstates the deployment and is reported only because it is the convention.

Table 9: The 95% bootstrap interval for annualised return, repeated across every candidate block length.

Block length	Lower	Upper	Contains zero
1	-47.01%	53.29%	yes
2	-46.03%	50.55%	yes
3	-45.41%	48.56%	yes
5	-45.60%	48.28%	yes
8	-47.09%	46.13%	yes
13	-49.26%	48.93%	yes
21	-47.01%	43.24%	yes

6.4 The conclusion does not depend on the arbitrary choices

A bootstrap block length is an arbitrary choice, so all seven candidates are reported rather than one (Table 9, Figure 9). Every interval contains zero and their widths vary by about ten percentage points out of ninety, so the interval is a statement about the data rather than about the block length. The automatic Politis–White selection lands at 2.12 sessions for the portfolio return series and at 12.26 for the loss differentials, which is why the two bootstraps in this paper carry different block lengths.

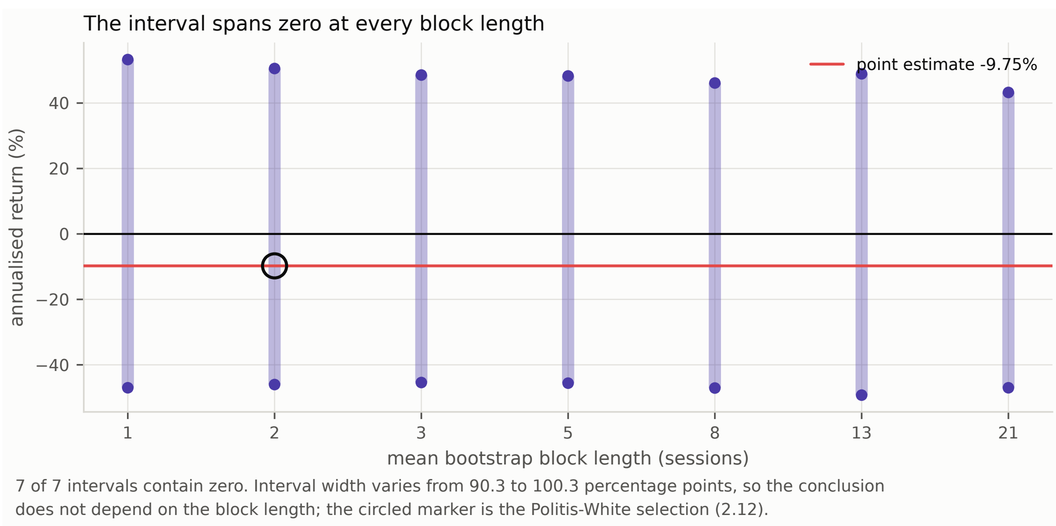

Figure 9: The 95% bootstrap interval for annualised return across seven block lengths. Every interval contains zero.

6.5 The design was never powered for a plausible effect

Table 10 inverts the tests that were actually run. At 120 sessions, the best-powered candidate could have been separated from the null only by an out-of-sample R_0^2 above 0.1132. That is roughly eleven times the upper end of what Campbell and Thompson [18] and Gu et al. [19] treat as economically meaningful, and about a hundred times the values those studies typically report. Detecting $R_0^2 = 0.01$ at the same size and power would need $15{,}126$ sessions of this panel: about sixty years of trading.

The portfolio side is worse. Because the search threshold does not shrink with the record length, establishing skill at the conventional deflated-Sharpe bar of 0.95 would have required a per-session Sharpe ratio of 0.2884, an annualised 4.5785. No equity strategy operates there. The correct reading of the deflated Sharpe ratio of 0.0452 is therefore not “the strategy has no skill” but “a 120-session record searched over 72 configurations cannot demonstrate skill at any plausible level, and this one does not come close either”.

Table 10: Inverting the accepted tests: the smallest effects this design could have separated from the null at conventional size and power.

Quantity	Value
Best-powered candidate	logistic
Standard error of its session-mean loss differential	1.444e-05
Test size α	0.05
Target power	0.80
Smallest detectable loss differential	4.079e-05
Mean squared error of the null	3.604e-04
Smallest detectable out-of-sample R_0^2	0.1132
Reference effect assumed plausible	0.0100
Sessions required for that effect	15,126
Per-session Sharpe required for a deflated ratio of 0.95	0.2884
Its annualised equivalent	4.5785

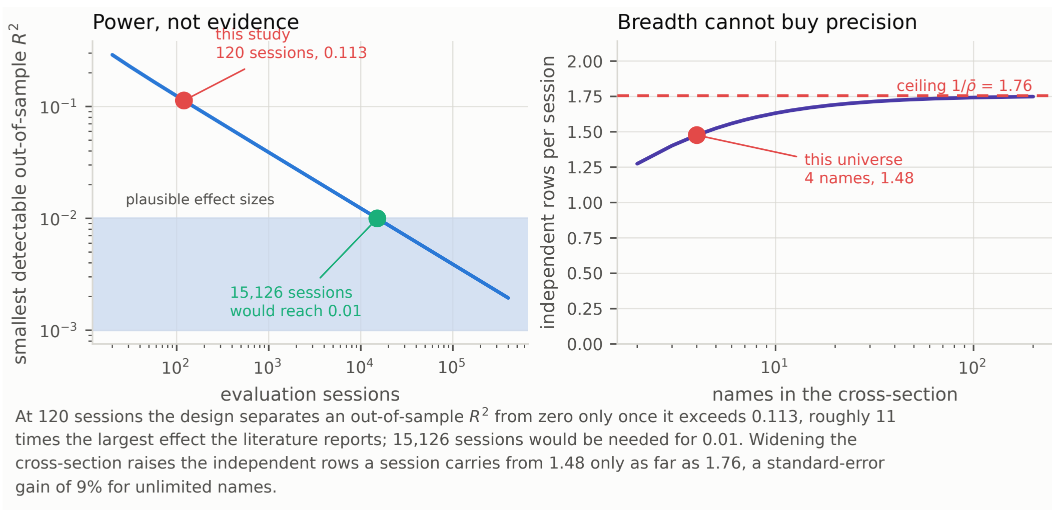

Figure 10: Left: the smallest out-of-sample R^2 separable from zero at 5% size and 80% power, as a function of the number of evaluation sessions, with this study marked and the band of effects the literature reports shaded. Right: independent rows per session against the number of names, with the $1/\bar{\rho}$ ceiling of Proposition 1.

The right panel of Figure 10 closes the obvious escape route. Proposition 1 caps the independent rows a session can carry at $1/\bar{\rho} = 1.7554$; this four-name panel already reaches 1.4765 of that. An unlimited universe at the same correlation would improve standard errors by 9%—which moves the detectable R_0^2 from 0.113 to about 0.095, not to 0.01. Breadth is not the missing ingredient. Sessions are.

6.6 The cost schedule rules the design out before any model is chosen

Evaluating Equation 12 on the accepted design gives the sharpest result in this paper (Table 11). At a round-trip cost of 20.2 basis points against a target standard deviation of 1.90%, holding the top 3 of 4 names—a selection score of only $\lambda(4, 3) = 0.3431$, since holding three quarters of the universe barely selects at all—requires a cross-sectional information coefficient of 0.3098. The ridge model achieved 0.0387. The design demanded eight times the skill it got, and the level it demanded is roughly an order of magnitude above what cross-sectional equity forecasting attains anywhere.

Figure 11 shows that this cannot be fixed by widening the universe alone. A wider ranking does lower the requirement, but $\lambda(N, k)$ converges to the mean of the corresponding tail of the normal, so past a few dozen names only a more selective rule helps: holding the top 2% of 500 names still

Table 11: Proposition 2 evaluated on the accepted design. The requirement depends on the cost schedule, the volatility of the target and the breadth of the rule, and on nothing the forecaster does.

Quantity	Value
Round-trip cost on notional	20.20 bp
Standard deviation of the executable target	1.90%
Names ranked	4
Names held	3
Selection score $\lambda(N, k)$	0.3431
Information coefficient required	0.3098
Information coefficient achieved	0.0387
Shortfall factor	8.00×

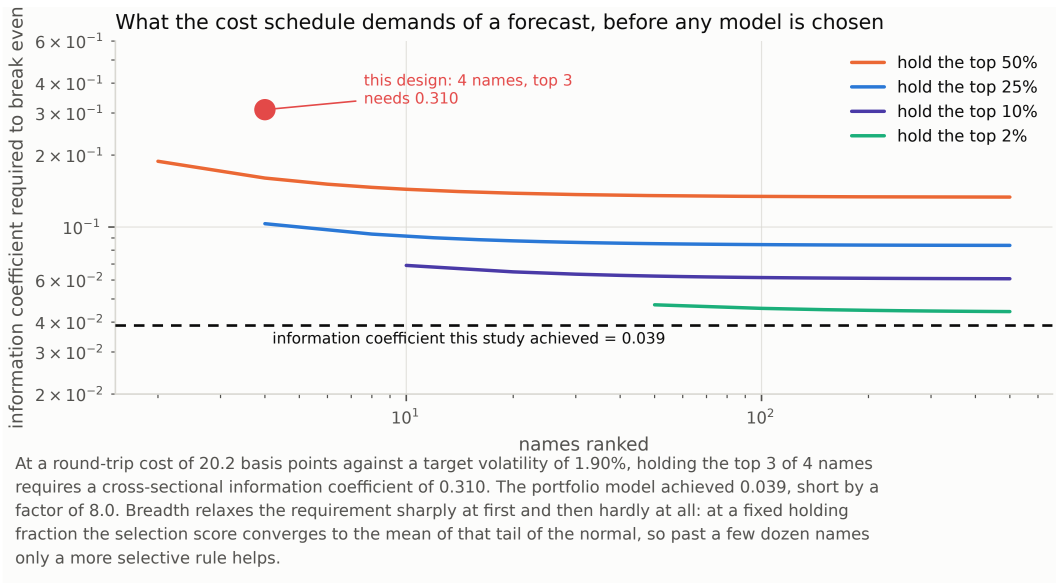

Figure 11: Proposition 2 evaluated across universe widths and holding fractions at this run's cost schedule. The information coefficient the design demands is compared against the one the portfolio model achieved.

requires an information coefficient of about 0.043, above what this study achieved. Under this cost schedule, at this target volatility, no long-only top-k rule fed by a forecast of this quality is profitable at any breadth.

This is worth separating from the statistical result, because the two are independent. The tests in Section 6.2 say the forecast is not distinguishable from zero. Proposition 2 says that even a forecast that *was* distinguishable from zero—at, say, an information coefficient of 0.05, which would be a respectable result in this literature—would still have lost money in this design. A study that reported such a forecast as a success, without evaluating the bound, would be reporting a statistically real and economically worthless effect.

6.7 Both readings of the search correction agree

The 72 configurations disperse considerably less than 72 independent trials would: their realised Sharpe variance is 0.002757 against the 0.008337 a single 120-session estimate carries. Solving Equation 13 puts $N_{\text{eff}} = 7.01$ (Table 12). The grid, in other words, asked about seven genuinely different things while appearing to ask seventy-two. That is a useful diagnostic in its own right: a search whose effective count is far below its nominal count is exploring one design repeatedly rather than exploring a design space.

Table 12: The False Strategy threshold under both readings of the grid. The best configuration falls below either, so the conclusion does not turn on how the trials are counted.

Quantity	Value
Configurations searched	72
Realised variance of trial Sharpe ratios	0.002757
Variance implied by independent trials	0.008337
Effective independent trials	7.01
Threshold at the realised dispersion	0.1267
Threshold if the trials were independent	0.2203
Best configuration in the grid	0.0356

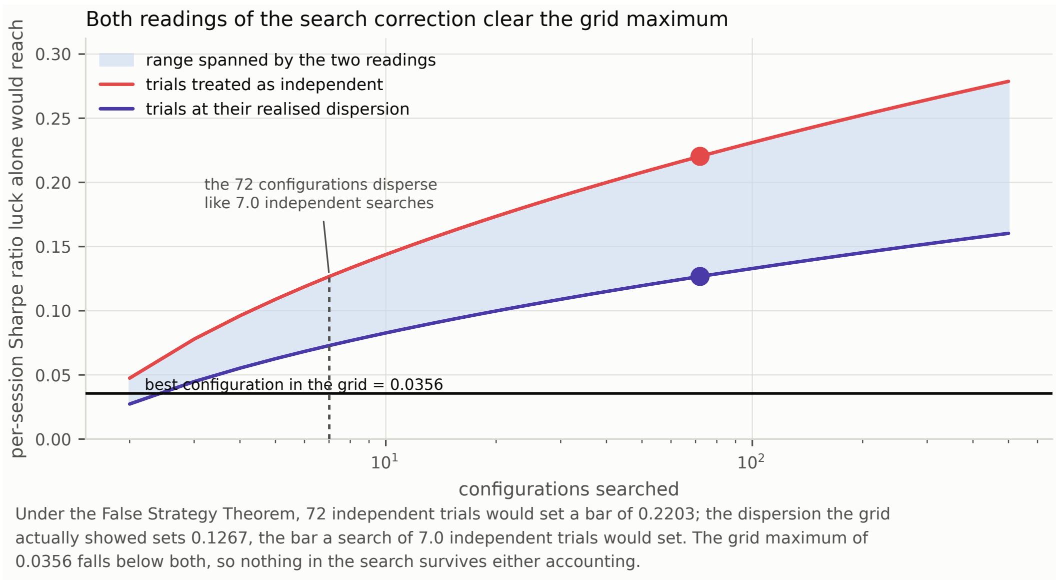

Figure 12: The False Strategy threshold as a function of the number of configurations searched, under independent trials and at the dispersion the grid actually showed. The vertical marker is N_{eff} from Proposition 3.

For the conclusion it matters less than one might expect, which is the point of reporting both. Treating the trials as independent sets the bar at 0.2203; using their realised dispersion sets it at 0.1267. The grid maximum is 0.0356, below both by a wide margin (Figure 12). Whatever one believes about how to count a correlated search, nothing in this grid survives it.

7 Discussion

7.1 What this result is, and is not

On this universe, over this window, at this horizon, scale-normalised price and volume features carry no out-of-sample information about next-session open-to-close returns that survives an honest accounting of sample size and search. The apparatus is not merely failing to reject: several models are measurably worse than the null, and the family is far from rejection jointly.

It does not follow that Borsa Istanbul is efficient, that no short-horizon signal exists, or that machine learning cannot forecast equity returns. It does not even follow that this particular signal is absent. Section 6.5 makes the honest version of the claim available for the first time: an effect large enough to be *usable* at this horizon on this universe would have shown up, and did not; an effect of the size the literature actually reports would not have shown up either way, and the experiment is silent about it.

Two details deserve emphasis. First, a model with no signal does not merely fail to help; it adds estimation variance to the forecast, which is a pure cost in a squared-error comparison against a zero forecast. Observing several fitted models below zero is therefore a consistency check rather than a surprise. Second, the False Strategy threshold is an *expectation*, not a critical value: a genuinely skill-free grid exceeds it about half the time, which the accompanying test suite verifies by simulation. It is reported as a diagnostic, and the deflated Sharpe ratio, which converts it into a probability, is the actual test.

7.2 What follows for the design of such studies

The bounds in Section 5 are cheap to compute and, we would argue, belong in the design phase rather than the post-mortem. Four of them can be evaluated before any data is modelled.

Compute the detectable effect before running the experiment. Equation 9 needs only the target's variance and the intended sample size. A design whose detectable effect exceeds the plausible effect by an order of magnitude will produce an uninformative answer whichever way it comes out, and should be lengthened or abandoned rather than run.

Compute the feasibility bound before choosing a model family. Equation 12 needs only the cost schedule, the target volatility and the intended breadth. It is the difference between "our model was not good enough" and "no model would have been", and in this study it is decisive. Reporting it also disciplines the reader: a paper that clears the statistical bar but not the feasibility bound has found something true and useless, and should say so.

State the unit of inference and defend it. The gap between the row-level and session-level p-values here is a median factor of 52. A panel study that does not say which it used has not reported its result.

Report the effective size of the search, not only its nominal size. $N_{\text{eff}} = 7.01$ against a nominal 72 says the grid was narrower than it looked. The opposite case—a search whose configurations are genuinely independent—deserves a correspondingly harsher correction, and neither is visible from the trial count alone.

7.3 Verifying the evaluator, not only the model

An evaluation apparatus is software, and software with no adversarial test is software of unknown quality. Sixteen real defects in the inference and reporting code—an autocovariance normalised by $n - k$ instead of n, a Holm correction without its running maximum, a Diebold–Mariano test aggregating rows instead of sessions, a deflated Sharpe ratio that ignores the trial count, and twelve others—are reintroduced one at a time by a harness that requires a named test to fail for each and to pass again once the edit is reverted (Appendix F). Three of the guarding tests turned out to be decorative and were repaired as a result.

We think this is the right standard for statistical code that produces published numbers, and it is cheaper than it sounds: the catalogue is a list of edits and test identifiers, and the harness is under a hundred lines. A negative result in particular has no external check—nobody notices when a broken test fails to find an effect that is not there—so the apparatus has to be checked directly.

8 Limitations

Four stocks and 120 evaluated sessions are too few for a strong economic conclusion, and Section 6.5 quantifies exactly how few. The universe is a fixed prototype rather than point-in-time index membership. The committed input contains no BIST index series, so only cash and an equal-weight eligible-universe benchmark are reported and no index-relative claim is made. Sector metadata is unavailable and is reported as unavailable rather than filled with a placeholder. The strategy holds risk on 62 of 120 sessions, so annualised figures assume a deployment that did not occur. The search grid is itself a choice: a different grid would give a different deflation threshold, and the grid is declared in the run configuration so the choice is auditable. Corporate-action transition and fail-closed policies are exercised by synthetic tests; the empirical snapshot contains cash dividends only.

The detectability results carry their own assumptions. Proposition 1 uses an equicorrelated approximation, which is a simplification of a correlation matrix whose realised pairwise values here range from 0.46 to 0.83; the ceiling it gives is an average-case statement. Proposition 2 assumes joint normality and an unbiased forecast, and ignores estimation error in the ranking—all of which make the bound optimistic, so a design that fails it fails a fortiori. The power calculation of Equation 9 treats the loss-differential standard error as fixed while the sample size varies, which is exact only if the higher moments of the differential are stable across window lengths.

Finally, the repository implements Kalman filters, Ornstein–Uhlenbeck mean reversion, GARCH, HMM regimes, wavelets, cointegration, Kelly sizing, gradient boosting, stacking, calibration and neural sequence models, none of which is part of the accepted experiment; their presence is not evidence of predictive value, and a machine-readable component status table records for each one whether it is integrated, evaluated, and backed by run evidence.

9 Conclusion

Seven forecasters were evaluated on a leakage-controlled Borsa Istanbul panel with an executable target, under tests that account for cross-sectional dependence, the number of models compared, and the number of configurations examined. No model beats the zero-return null, four are significantly worse, the joint data-snooping null is not rejected at any conventional level, and the best of 72 configurations does not reach the Sharpe ratio that skill-free search alone would produce. The reported strategy loses money after costs and would lose money at $0.62\times$ the modelled cost schedule.

Inverting the same apparatus turns that from a verdict about the market into a verdict about the experiment. The design could only have resolved an out-of-sample R^2 above 0.113, roughly eleven times the largest effect this literature credibly reports; reaching 0.01 would take $15{,}126$ sessions. Widening the cross-section cannot substitute, because a session of correlated names carries at most $1/\bar{\rho}$ independent observations. And the cost schedule demanded an information coefficient of 0.310 from a design that delivered 0.039—a bound that no choice of model family could have met, and that would have been known before any model was fitted had it been computed.

The useful output is therefore the apparatus and its self-assessment: a target the backtest can trade, an evaluation whose unit of inference matches the data's dependence structure, a search that is counted rather than hidden, an explicit statement of what the design could and could not have found, and a run that replays byte-for-byte. The next legitimate step is a longer point-in-time dated universe and a cheaper execution assumption, chosen so that the feasibility bound is cleared before the first model is fitted—not another model family.

Reproducibility

Every table and figure in this manuscript is generated from one immutable run bundle by a tested module and regenerated by `make paper`; the manuscript cannot state a table entry the pipeline does not produce. The run replays byte-for-byte from bundled inputs with `make reproduce-committed`, `make verify-claims` checks the prose figures in the repository's README against the same artifacts, and `make mutation-check` reintroduces sixteen real defects into the source and requires the guarding test to fail for each. Code and artifacts: https://github.com/ITheClixs/Prediction-System-for-Istanbul-Stock-Exchange.

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A Proofs

Proof of Proposition 1. Let $x_1, \dots, x_k$ be the session's targets, each of variance σ^2, with $\operatorname{corr}(x_i, x_j) = \bar\rho$ for $i \neq j$. The variance of their mean is

$$\operatorname{Var}\left(\frac{1}{k}\sum_i x_i\right) = \frac{\sigma^2}{k^2}\left(k + k(k-1)\bar\rho\right) = \frac{\sigma^2}{k}\left(1 + (k-1)\bar\rho\right), \tag{14}$$

so the session mean is as informative as $m(k) = k/(1 + (k-1)\bar\rho)$ independent observations. Differentiating,

$$m'(k) = \frac{1-\bar\rho}{\left(1+(k-1)\bar\rho\right)^2} > 0 \quad \text{for } \bar\rho < 1, \tag{15}$$

so m is strictly increasing, and dividing numerator and denominator by k gives $m(k) \to 1/\bar\rho$ as $k \to \infty$. Since m is increasing and convergent, the limit is a supremum. □

Proof of Proposition 2. Write $\hat r$ in standardised form $z = (\hat r - \mu_{\hat r})/\sigma_{\hat r}$. For a bivariate normal pair with correlation ρ,

$$\mathbb{E}[r \mid z] = \rho\,\sigma\,z. \tag{16}$$

Selection holds the k names with the largest $\hat r$, which is a function of the standardised scores alone; conditional on the ranking, the selected names have standardised scores distributed as the top k order statistics of N standard normal draws. By the tower property the expected realised return of an equally weighted holding is

$$\mathbb{E}\left[\frac{1}{k}\sum_{i=N-k+1}^{N} r_{(i)}\right] = \rho\,\sigma\,\frac{1}{k}\sum_{i=N-k+1}^{N}\mathbb{E}\big[Z_{(i:N)}\big] = \rho\,\sigma\,\lambda(N,k). \tag{17}$$

Charging the round-trip cost c on the notional of each position, the expected net return per name is $\rho\,\sigma\,\lambda(N,k) - c$, and requiring it to be positive gives Equation 12. The second form of λ in Equation 11 follows from exchangeability: a given draw lies among the top k exactly when at most $k-1$ of the remaining $N-1$ draws exceed it, so

$$\sum_{i=N-k+1}^{N}\mathbb{E}\big[Z_{(i:N)}\big] = N\,\mathbb{E}[Z\cdot\mathbf{1}\{Z \text{ in top } k\}] = N\int_0^1 \Phi^{-1}(u)\,\Pr[\operatorname{Bin}(N-1,\,1-u) \le k-1]\,\mathrm{d}u, \tag{18}$$

and the binomial tail is the regularised incomplete beta function $I_u(N-k,\,k)$. □

Proof of Proposition 3. Φ^{-1} is strictly increasing, and $1 - 1/N$ and $1 - 1/(Ne)$ are both strictly increasing in N on $(1, \infty)$; q is a convex combination of the two with positive weights $1 - \gamma$ and γ, hence strictly increasing and continuous there, with $q(N) \to -\infty$ as $N \downarrow 1$ and $q(N) \to \infty$ as $N \to \infty$. The right-hand side of Equation 13 is therefore a continuous strictly increasing bijection from $(1, \infty)$ onto $\mathbb{R}$, so for any fixed left-hand side the equation has exactly one solution. Bisection converges to it. □

Remark 2. Proposition 3 identifies N_{eff} by matching expected maxima, not by matching the full distribution of the maximum. Two searches with the same expected maximum can differ in the tail, so N_{eff} should be read as a summary of how much independent exploration the grid represents rather than as a claim that the grid is distributionally equivalent to N_{eff} independent trials.

B Feature manifest

Table 13: The accepted feature manifest, schema version `2.0.0`, digest `b0349ccf178ddaf1`. Every entry declares its formula, its lookback in sessions, and a normalisation policy of `none`: the accepted models consume scale-free quantities, so nominal price level cannot act as accidental ticker identification. A panel built against a different manifest digest is refused.

#	Feature	Formula	Lookback
1	`log_return_1d`	`log(split_close_t / split_close_t_minus_1)`	2
2	`log_return_5d`	`log(split_close_t / split_close_t_minus_5)`	6
3	`log_return_20d`	`log(split_close_t / split_close_t_minus_20)`	21
4	`close_over_sma20_minus_1`	`split_close / mean_20(split_close) - 1`	20
5	`sma20_over_sma100_minus_1`	`mean_20(split_close) / mean_100(split_close) - 1`	100
6	`atr14_over_close`	`atr_14(split_adjusted_ohlc) / split_close`	15
7	`vwap20_over_close_minus_1`	`vwap_20(split_adjusted_hlc, raw_volume) / split_close - 1`	20
8	`log_volume`	`log1p(raw_volume)`	1
9	`volume_zscore_20`	`zscore_20(raw_volume)`	20
10	`realized_volatility_20`	`std_20(log_return_1d) * sqrt(252)`	21
11	`intraday_range_over_close`	`(split_high - split_low) / split_close`	1
12	`overnight_gap`	`split_open / prior_split_close - 1`	2
13	`drawdown_20`	`split_close / max_20(split_close) - 1`	20
14	`cross_sectional_return_rank`	`midrank_percentile_by_date(log_return_20d)`	21
15	`market_relative_return_20d`	`log_return_20d - date_mean(log_return_20d)`	21
16	`day_of_week_sin`	`sin(2*pi*weekday/7)`	1
17	`day_of_week_cos`	`cos(2*pi*weekday/7)`	1
18	`month_sin`	`sin(2*pi*(month-1)/12)`	1
19	`month_cos`	`cos(2*pi*(month-1)/12)`	1

C Run configuration

Table 14: The committed run configuration, verbatim from the bundle. These values enter the configuration hash, so a change to any of them produces a different run identity rather than a silently different result.

Parameter	Value
Scope	
`experiment_scope`	`fixed_bist_large_cap_prototype`
`methodology_version`	`accepted-baseline-v2`
`seed`	`42`
Fold geometry	
`min_train_dates`	`24`
`validation_dates`	`10`
`step_dates`	`10`
`embargo_dates`	`1`
Portfolio	
`portfolio_model`	`ridge`
`top_k`	`3`
`starting_equity`	`100000`

Parameter	Value
`min_trade_value`	100
`max_participation`	0.01
`liquidity_lookback_sessions`	20
`decision_cost_rate`	0.0001
Cost schedule	
`commission_rate`	0.0002
`bid_ask_spread_rate`	0.001
`slippage_rate`	0.0003
`market_impact_coefficient`	0.0001
`tax_rate`	0
Resampling and search	
`bootstrap_iterations`	10000
`bootstrap_block_sizes`	1,2,3,5,8,13,21
`sensitivity_grid`	train=24,36,48|val=5,10,20|embargo=1,2|topk=1,2,3,4

D Executed folds

Table 15: The partition that was executed, read from the run's fold artifact rather than recomputed for the manuscript. Training windows expand from a common start; the embargo column counts trading dates removed between training and validation; validation blocks are disjoint, so no session is evaluated twice.

Fold	Train from	Train to	Dates	Emb.	Validate from	Validate to	Dates
fold_0001	2025-09-01	2025-10-02	24	1	2025-10-06	2025-10-17	10
fold_0002	2025-09-01	2025-10-16	34	1	2025-10-20	2025-11-03	10
fold_0003	2025-09-01	2025-10-31	44	1	2025-11-04	2025-11-17	10
fold_0004	2025-09-01	2025-11-14	54	1	2025-11-18	2025-12-01	10
fold_0005	2025-09-01	2025-11-28	64	1	2025-12-02	2025-12-15	10
fold_0006	2025-09-01	2025-12-12	74	1	2025-12-16	2025-12-29	10
fold_0007	2025-09-01	2025-12-26	84	1	2025-12-30	2026-01-13	10
fold_0008	2025-09-01	2026-01-12	94	1	2026-01-14	2026-01-27	10
fold_0009	2025-09-01	2026-01-26	104	1	2026-01-28	2026-02-10	10
fold_0010	2025-09-01	2026-02-09	114	1	2026-02-11	2026-02-24	10
fold_0011	2025-09-01	2026-02-23	124	1	2026-02-25	2026-03-10	10
fold_0012	2025-09-01	2026-03-09	134	1	2026-03-11	2026-03-25	10

E Full configuration grid

Table 16: Every configuration in the search grid, ordered by per-session Sharpe ratio. Each row is a complete re-run of the evaluation under one fold geometry and one portfolio breadth, not a re-weighting of a cached result. The final column names the model with the best out-of-sample R_0^2 in that trial.

Configuration	Folds	Sessions	Net return	Sharpe	Round trips	Best model
train24_val5_step5_emb2_k3	25	125	4.63%	0.0356	155	zero_return
train24_val5_step5_emb2_k1	25	125	4.28%	0.0304	67	zero_return
train24_val10_step10_emb2_k1	12	120	1.67%	0.0169	65	zero_return
train24_val5_step5_emb2_k2	25	125	1.62%	0.0163	115	zero_return
train24_val5_step5_emb2_k4	25	125	-0.19%	0.0049	189	zero_return
train36_val20_step20_emb1_k1	5	100	-0.67%	0.0013	49	zero_return
train36_val20_step20_emb2_k1	5	100	-1.54%	-0.0054	49	zero_return
train24_val10_step10_emb2_k3	12	120	-1.69%	-0.0060	150	zero_return
train24_val5_step5_emb1_k3	25	125	-2.36%	-0.0092	146	zero_return
train48_val20_step20_emb2_k1	5	100	-2.68%	-0.0102	69	zero_return
train48_val10_step10_emb2_k2	10	100	-2.53%	-0.0113	122	zero_return

Configuration	Folds	Sessions	Net return	Sharpe	Round trips	Best model
train48_val20_step20_emb2_k2	5	100	-3.05%	-0.0161	131	zero_return
train24_val10_step10_emb1_k1	12	120	-3.58%	-0.0162	62	zero_return
train24_val5_step5_emb1_k2	25	125	-4.54%	-0.0211	109	zero_return
train48_val20_step20_emb1_k1	5	100	-5.16%	-0.0282	68	zero_return
train24_val10_step10_emb1_k3	12	120	-4.77%	-0.0284	143	zero_return
train36_val10_step10_emb2_k2	11	110	-5.41%	-0.0315	121	zero_return
train24_val5_step5_emb1_k1	25	125	-6.99%	-0.0343	64	zero_return
train24_val20_step20_emb2_k1	6	120	-7.36%	-0.0371	73	zero_return
train24_val5_step5_emb1_k4	25	125	-6.52%	-0.0372	180	zero_return
train24_val10_step10_emb2_k2	12	120	-6.57%	-0.0380	111	zero_return
train24_val10_step10_emb2_k4	12	120	-6.14%	-0.0388	183	zero_return
train36_val10_step10_emb2_k1	11	110	-8.37%	-0.0488	66	zero_return
train24_val20_step20_emb2_k3	6	120	-8.20%	-0.0492	178	zero_return
train24_val20_step20_emb1_k1	6	120	-9.57%	-0.0507	71	zero_return
train48_val20_step20_emb1_k2	5	100	-7.28%	-0.0512	129	zero_return
train36_val20_step20_emb1_k2	5	100	-7.10%	-0.0528	85	zero_return
train48_val10_step10_emb2_k1	10	100	-8.86%	-0.0529	65	zero_return
train24_val10_step10_emb1_k4	12	120	-8.59%	-0.0575	174	zero_return
train48_val5_step5_emb2_k2	20	100	-8.49%	-0.0577	115	zero_return
train48_val10_step10_emb1_k2	10	100	-8.35%	-0.0579	118	zero_return
train36_val20_step20_emb2_k2	5	100	-7.77%	-0.0586	86	zero_return
train36_val10_step10_emb1_k2	11	110	-8.96%	-0.0591	121	zero_return
train36_val10_step10_emb1_k1	11	110	-9.82%	-0.0592	65	zero_return
train24_val10_step10_emb1_k2	12	120	-9.62%	-0.0599	106	zero_return
train24_val20_step20_emb2_k2	6	120	-10.81%	-0.0667	128	zero_return
train48_val10_step10_emb1_k1	10	100	-10.73%	-0.0697	62	zero_return
train48_val20_step20_emb2_k3	5	100	-10.10%	-0.0713	187	zero_return
train24_val20_step20_emb1_k3	6	120	-11.29%	-0.0718	172	zero_return
train48_val5_step5_emb2_k3	20	100	-9.74%	-0.0723	153	zero_return
train36_val5_step5_emb2_k2	22	110	-11.47%	-0.0744	111	zero_return
train24_val20_step20_emb2_k4	6	120	-12.57%	-0.0829	222	zero_return
train36_val10_step10_emb1_k3	11	110	-11.47%	-0.0843	170	zero_return
train36_val20_step20_emb1_k3	5	100	-9.90%	-0.0864	114	zero_return
train36_val5_step5_emb1_k2	22	110	-13.05%	-0.0871	108	zero_return
train36_val5_step5_emb2_k1	22	110	-14.34%	-0.0900	62	zero_return
train24_val20_step20_emb1_k2	6	120	-14.07%	-0.0914	124	zero_return
train36_val10_step10_emb2_k3	11	110	-12.22%	-0.0916	168	zero_return
train36_val20_step20_emb2_k3	5	100	-10.70%	-0.0945	116	zero_return
train48_val20_step20_emb1_k3	5	100	-12.64%	-0.0956	184	zero_return
train24_val20_step20_emb1_k4	6	120	-14.96%	-0.1016	214	zero_return
train48_val10_step10_emb2_k3	10	100	-13.98%	-0.1075	167	zero_return
train48_val5_step5_emb2_k4	20	100	-13.87%	-0.1086	186	zero_return
train48_val5_step5_emb2_k1	20	100	-15.86%	-0.1097	64	zero_return
train36_val5_step5_emb2_k3	22	110	-15.51%	-0.1152	153	zero_return
train36_val5_step5_emb1_k3	22	110	-16.25%	-0.1213	150	zero_return
train48_val20_step20_emb2_k4	5	100	-15.38%	-0.1226	238	zero_return
train36_val5_step5_emb1_k1	22	110	-19.14%	-0.1241	60	zero_return
train48_val5_step5_emb1_k1	20	100	-17.47%	-0.1266	61	zero_return
train36_val10_step10_emb1_k4	11	110	-17.02%	-0.1317	212	zero_return
train36_val20_step20_emb1_k4	5	100	-14.34%	-0.1353	141	zero_return
train48_val5_step5_emb1_k2	20	100	-18.46%	-0.1407	110	zero_return
train48_val10_step10_emb2_k4	10	100	-17.83%	-0.1430	207	zero_return
train36_val20_step20_emb2_k4	5	100	-15.04%	-0.1433	143	zero_return
train48_val20_step20_emb1_k4	5	100	-17.33%	-0.1436	234	zero_return
train48_val10_step10_emb1_k3	10	100	-16.85%	-0.1438	163	zero_return
train36_val10_step10_emb2_k4	11	110	-18.22%	-0.1445	209	zero_return
train48_val5_step5_emb1_k3	20	100	-17.77%	-0.1449	149	zero_return
train36_val5_step5_emb2_k4	22	110	-19.32%	-0.1501	184	zero_return
train36_val5_step5_emb1_k4	22	110	-20.13%	-0.1576	180	zero_return
train48_val5_step5_emb1_k4	20	100	-21.55%	-0.1829	181	zero_return
train48_val10_step10_emb1_k4	10	100	-20.98%	-0.1852	202	zero_return

F Mutation catalogue

Table 17: The 16 defects the mutation harness reintroduces into the source. For each one it runs the guarding test clean, applies the edit, requires the test to fail, restores the file, and requires the test to pass again. A defect that survives means the guarding test is decorative; three were found that way and repaired.

#	Subject	Defect reintroduced
1	variance guard	compare dispersion against zero instead of the series scale
2	variance guard (sharpe module)	same defect on the per-period estimator
3	holding period	hardcode one session instead of reading the ledger
4	Holm	drop the running maximum that enforces monotonicity
5	Diebold-Mariano	drop the Harvey-Leybourne-Newbold small-sample correction
6	Diebold-Mariano	aggregate rows instead of sessions, ignoring cross-sectional dependence
7	Diebold-Mariano	guard the differential variance against zero, not against scale
8	HAC	normalise the autocovariance by n-k instead of n
9	SPA	compare floored maxima, collapsing the comparison onto the atom at zero
10	SPA	skip the bootstrap recentring entirely
11	Lo annualisation	sum q-1 autocorrelations from a 120-session sample
12	Deflated Sharpe	ignore the trial count and never deflate
13	Cross-sectional dependence	assume independence by ignoring the correlation
14	Stationary bootstrap	use fixed-length blocks instead of geometric ones
15	Sensitivity grid	decouple the step from the fold width
16	Accepted config	allow a grid that omits the reported configuration